CHAPTER THREE: RESEARCH METHODOLOGY

Predicting Maternal Health Risk Among Pregnant Women in Kenya Using Machine Learning and Bayesian Inference

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3.1 Introduction

This chapter describes the research methodology adopted to achieve the three objectives of this study: (1) the development of a machine learning classification model for individual maternal health risk prediction using the Kenya Demographic and Health Survey 2022 (KDHS 2022) dataset; (2) the integration of Bayesian inference for probabilistic uncertainty quantification; and (3) the deployment of an AI-powered, Claude-integrated decision support application for Community Health Workers (CHWs) in Kenya. The chapter is structured around the research design, data collection methods, experimental procedures, data analysis methods, validation and evaluation framework, and ethical considerations governing the study. All methodological decisions are grounded in current literature and guided by the principle of reproducibility, transparency, and clinical relevance.

3.2 Design and Experimental Methodology

This study employs a quantitative, computational experimental methodology rooted in the data science research paradigm. The methodological framework follows a supervised machine learning pipeline, structured as a five-phase experimental design: (1) data acquisition and preprocessing, (2) feature engineering and target variable construction, (3) model development and comparative evaluation, (4) Bayesian inference integration, and (5) application deployment and validation. This pipeline is consistent with the CRISP-DM (Cross-Industry Standard Process for Data Mining) framework, which is widely adopted in applied data science research for health informatics (Wirth & Hipp, 2000).

The study adopts a comparative experimental design, evaluating multiple machine learning architectures against one another using standardised evaluation metrics. This design enables rigorous identification of the best-performing model for the specific characteristics of the KDHS 2022 tabular dataset, including its high dimensionality, categorical feature dominance, and class imbalance. Novel deep learning architectures for tabular data — specifically TabNet (Arik & Pfister, 2021) and TabTransformer (Huang et al., 2020) — are evaluated alongside classical ensemble methods, reflecting the current frontier of health ML research (Doshi et al., 2025; Sumon et al., 2025).

3.3 Research Design

This study adopts a non-experimental, quantitative secondary data analysis research design. The design is classified as non-experimental because no primary data collection, manipulation of variables, or randomised controlled trial is involved. Instead, the study analyses existing survey microdata from the KDHS 2022 — a nationally representative cross-sectional survey — to develop and validate a predictive model. The cross-sectional nature of the KDHS data means that risk prediction is based on a single point-in-time snapshot of each woman's demographic, socioeconomic, and reproductive health profile, which is appropriate for the study's objective of generating a community-level risk screening tool rather than a longitudinal clinical prognosis model.

The ontological position of this study is post-positivist: the research assumes that an objective maternal health risk exists and that it can be measured and predicted with quantifiable accuracy, while acknowledging that all models are imperfect approximations of reality, which is why Bayesian uncertainty quantification is incorporated as a core component rather than an afterthought. The epistemological stance is empiricist — knowledge about maternal risk is derived from observation (survey data) and validated through systematic model evaluation rather than expert elicitation alone.

Design Component	Specification
Research Paradigm	Post-positivist, quantitative
Research Design	Non-experimental, secondary data analysis
Data Type	Cross-sectional survey microdata (KDHS 2022)
Unit of Analysis	Individual woman of reproductive age (15–49 years)
Target Population	Pregnant women or women with a recent birth in Kenya
Analytical Approach	Supervised machine learning classification + Bayesian inference
Geographic Scope	National (Kenya, 47 counties)
Temporal Scope	KDHS 2022 survey period (2022–2023)
End Product	Streamlit web application with Claude AI integration

3.4 Data Collection Methods

3.4.1 Primary Data Source

This study uses the Kenya Demographic and Health Survey 2022 (KDHS 2022) as its sole primary data source. The KDHS is a nationally representative household survey conducted by the Kenya National Bureau of Statistics (KNBS) in partnership with ICF International, with technical support from the DHS Program funded by the United States Agency for International Development (USAID). The 2022 survey covered all 47 counties of Kenya using a stratified two-stage cluster sampling design, yielding a final sample of 17,414 women aged 15–49 years who completed the women's questionnaire. The Individual Recode (IR) file — which contains all individual-level women's data — was accessed through the DHS Program's data sharing portal (dhsprogram.com) following institutional registration and data use agreement approval.

3.4.2 Dataset Characteristics

Upon loading, the KDHS 2022 IR file contained 32,156 records across 5,925 variables, reflecting the inclusion of multiple birth-history records per woman. After filtering to retain only women with complete data for a recent birth (within the five years preceding the survey), the working dataset was reduced to 10,391 records. Thirteen variables were selected for the analysis based on clinical evidence, data availability, and relevance to the composite maternal risk outcome. The selected variables and their KDHS codes are presented in the following table:

KDHS Code	Variable Name	Clinical Justification
v012	**Maternal age**	Young (<18) and older (>35) mothers face elevated obstetric risk
v201	**Parity (total births)**	High parity (≥5) is associated with uterine atony and haemorrhage
v106	**Education level**	Lower education correlates with reduced health-seeking behaviour
v190	**Wealth index (1–5)**	Poverty is a primary barrier to skilled attendance and facility delivery
v025	**Urban/rural residence**	Rural residence limits access to emergency obstetric care
v501	**Marital status**	Social support influences antenatal care attendance
v212	**Age at first birth**	Early first birth (<18) signals compounded cumulative risk
v024	**Region/County**	County-level disparities in maternal outcomes are well documented
m14_1	**ANC visits (recent birth)**	Fewer than 4 ANC visits is a WHO-defined risk indicator
m15_1	**Place of delivery**	Home delivery significantly elevates complication risk
m17_1	**Caesarean section**	Prior CS affects subsequent delivery risk profile
m3a_1	**Pregnancy complication**	Self-reported complications are a direct risk indicator
b11_01	**Preceding birth interval**	Interval <24 months is associated with adverse maternal outcomes

3.5 Design and Experiment Procedures

3.5.1 Phase 1 — Data Preprocessing

All data preprocessing will be conducted in Python 3.11 using the pandas, NumPy, and scikit-learn libraries within a Google Colab environment. The preprocessing pipeline consists of the following sequential steps:

- Missing value treatment: v212 (age at first birth) contains 33 missing values — these will be imputed using the column median. b11_01 (birth interval) contains 2,835 missing values representing first-birth women — these will be imputed with 0, denoting no preceding birth.
- Categorical encoding: Ordinal variables (v106, v190, v025, v501, v024) will be encoded using Label Encoding. Binary variables (m15_1, m17_1, m3a_1) will be encoded as 0/1.
- Feature scaling: Continuous variables (v012, v201, v212, b11_01, m14_1) will be standardised using scikit-learn's StandardScaler to ensure uniform feature contribution to distance-based model components.
- Class imbalance handling: The composite target variable is expected to be imbalanced. SMOTE (Synthetic Minority Oversampling Technique) will be applied exclusively to the training set to prevent data leakage.
- Train-validation-test split: The dataset will be partitioned into 70% training, 15% validation, and 15% test sets using stratified splitting to preserve class proportions.

3.5.2 Phase 2 — Target Variable Engineering

The KDHS 2022 dataset does not contain a pre-existing maternal risk classification variable. A composite maternal risk score will be engineered using a clinically grounded rule-based system informed by WHO antenatal care guidelines and the reviewed literature. A woman is assigned a risk flag for each of the following conditions:

Risk Condition	KDHS Variable	Clinical Basis
Fewer than 4 ANC visits	m14_1 < 4	WHO minimum ANC standard
Home/non-facility delivery	m15_1 > 20	Absence of skilled birth attendance
Reported pregnancy complication	m3a_1 == 1	Direct obstetric risk indicator
Maternal age under 18	v012 < 18	Adolescent pregnancy risk
Maternal age over 35	v012 > 35	Advanced maternal age risk
Parity of 5 or more	v201 >= 5	Grand multiparity risk
Birth interval under 24 months	b11_01 < 24 & b11_01 > 0	Short inter-pregnancy interval
Wealth quintile 1 or 2	v190 <= 2	Poverty-related access barrier

The total number of flags per woman constitutes her raw risk score. This score is then classified into three ordinal risk levels: Low Risk (0 flags), Moderate Risk (1–2 flags), and High Risk (3 or more flags). This three-class classification target reflects clinical priority levels used by CHWs in Kenya's maternal health referral system.

3.5.3 Phase 3 — Model Development and Comparative Evaluation

Six machine learning models will be trained and evaluated in a progressive architecture comparison:

- **Tier 1 (Baseline):** Logistic Regression — baseline interpretable model for benchmarking
- **Tier 1 (Baseline):** Decision Tree Classifier — visual decision pathway baseline
- **Tier 2 (Classical Ensemble):** Random Forest Classifier — classical ensemble with built-in feature importance
- **Tier 2 (Classical Ensemble):** XGBoost Classifier — gradient boosting with regularisation
- **Tier 2 (Classical Ensemble):** CatBoost Classifier — optimised for categorical variables dominant in KDHS
- **Tier 3 (Novel Architecture):** TabNet — sequential attention deep learning architecture for tabular data (Arik & Pfister, 2021); validated on maternal data by Tariq et al. (2025, ScienceDirect)
- **Tier 3 (Novel Architecture):** TabTransformer — contextual embedding transformer for categorical features (Huang et al., 2020); applied to obstetric prediction by Mursil et al. (2025, IEEE JBHI)
- **Tier 4 (Final Model):** Stacking Ensemble — meta-learner (Logistic Regression) combining outputs of TabNet, XGBoost, and CatBoost

All models will be trained on the SMOTE-balanced training set. Hyperparameter optimisation for Tier 2–4 models will be performed using Optuna, a Bayesian hyperparameter optimisation framework, with 100 trials per model under 5-fold stratified cross-validation. The final test set will be used only once, after model selection is complete, to prevent information leakage.

3.5.4 Phase 4 — Bayesian Inference Integration

Following model selection, Bayesian inference will be applied using PyMC (version 5.x) to quantify predictive uncertainty. A Bayesian logistic regression model will be fitted with weakly informative priors on the coefficient parameters, generating posterior probability distributions for each risk class. The posterior means will be calibrated against the selected ML model's predict_proba() outputs using CalibratedClassifierCV to produce well-calibrated probability estimates. For each individual prediction, the application will display: the risk class (Low/Moderate/High), the posterior mean probability, and the 95% highest density interval (HDI) — providing CHWs with a confidence-aware risk score.

3.5.5 Phase 5 — Application Development

The final end product will be developed using Streamlit (Python), deployed on Streamlit Community Cloud, and integrated with the Anthropic Claude API (claude-sonnet-4-6 model). The application architecture consists of three layers: (1) the input layer — a CHW-facing form capturing the 13 patient variables; (2) the prediction layer — the trained model generates a risk classification and Bayesian probability score; (3) the AI explanation layer — SHAP values identifying the top three risk drivers are passed to Claude via API prompt, which returns a plain-language explanation, care recommendations, and a referral letter in the CHW's selected language (English or Swahili).

3.6 Data Analysis Methods

3.6.1 Exploratory Data Analysis (EDA)

Prior to modelling, comprehensive exploratory data analysis will be conducted to understand the distributional properties of the 13 selected variables and the target variable. EDA will include: univariate frequency distributions and descriptive statistics for all variables; bivariate analysis examining associations between each predictor and the target risk class using chi-square tests for categorical variables and ANOVA for continuous variables; correlation matrix and heatmap analysis to detect multicollinearity; and missing value profiling. EDA will be conducted using pandas, matplotlib, and seaborn.

3.6.2 Feature Importance Analysis

Feature importance will be assessed using two complementary approaches: (1) built-in feature importance from tree-based models (Random Forest, XGBoost, CatBoost) based on Gini impurity and gain; and (2) SHAP (SHapley Additive exPlanations) values computed using the shap Python library. SHAP provides model-agnostic, theoretically grounded feature attribution that satisfies local accuracy, missingness, and consistency axioms (Lundberg & Lee, 2017). Both global SHAP summary plots (ranking variables by mean absolute SHAP value across all predictions) and individual SHAP waterfall plots (explaining single predictions) will be generated. The TabNet model's built-in feature selection via sequential attention masks will also be analysed as a third source of interpretability, consistent with the approach validated by Tariq et al. (2025) for obstetric prediction.

3.6.3 Statistical Analysis

Descriptive statistics will be reported as means and standard deviations for continuous variables and frequencies and proportions for categorical variables. Inferential statistics will be used to validate the composite risk variable — chi-square goodness-of-fit tests will assess whether the risk class distribution departs significantly from a uniform distribution, and Spearman rank correlations will assess monotonic relationships between continuous predictors and the ordinal risk score. All statistical tests will be conducted at the 0.05 significance level.

3.7 Validation and Evaluation

3.7.1 Model Performance Metrics

Model performance will be evaluated using a comprehensive set of metrics appropriate for a multi-class imbalanced classification problem:

Metric	Definition	Justification
Accuracy	Proportion of correctly classified instances	Overall baseline performance indicator

Metric	Definition	Justification
Precision (macro)	Proportion of positive predictions that are correct, averaged across classes	Controls false referral rate
Recall / Sensitivity (macro)	Proportion of actual positive cases correctly identified	Critical in healthcare — missed High Risk cases are costliest errors
F1-Score (macro)	Harmonic mean of precision and recall	Balances precision-recall trade-off under class imbalance
AUC-ROC (macro OvR)	Area under the receiver operating characteristic curve	Measures discriminative ability across all class thresholds
Matthews Correlation Coefficient (MCC)	Balanced measure for multi-class imbalanced problems	Superior to accuracy for imbalanced datasets
Calibration (Brier Score)	Mean squared error of probability predictions	Validates quality of Bayesian probability outputs

> *Priority Metric: In healthcare risk prediction, Recall (Sensitivity) for the High Risk class is the primary performance criterion — a missed high-risk woman is a far more costly error than a false alarm. All model selection decisions will prioritise High Risk class recall alongside macro AUC-ROC.*

3.7.2 Cross-Validation Strategy

All model training and hyperparameter tuning will use 5-fold stratified cross-validation, ensuring that each fold preserves the class proportion of the target variable. The final model performance will be reported on the held-out 15% test set, which will not be used during training or tuning. This two-stage validation strategy (cross-validation for tuning, hold-out set for final evaluation) prevents overfitting and provides an unbiased estimate of generalisation performance.

3.7.3 Application Validation

The Streamlit application will undergo two rounds of validation: (1) technical validation — the application will be tested with 50 synthetic patient profiles spanning all risk levels to verify that predictions match the trained model's outputs; and (2) content validation — the Claude AI-generated explanations and referral letters will be reviewed by the study supervisor and at least one peer reviewer for clinical coherence, language clarity, and accuracy of risk driver attribution. Any systematic errors in Claude's outputs will be addressed by refining the API prompt template.

3.8 Ethical Considerations

3.8.1 NACOSTI and IRB Ethical Approval

This study will be submitted to the National Commission for Science, Technology and Innovation (NACOSTI) and the Institutional Review Board (IRB) of USIU-Africa for ethical review prior to the commencement of the implementation phase. Given that the study uses a fully anonymised

secondary dataset with no primary data collection involving human participants, it is classified as minimal risk research and an expedited review is anticipated.

3.8.2 Data Privacy and Confidentiality

- The KDHS 2022 dataset is fully de-identified by KNBS and ICF International prior to public release. No personally identifiable information (PII) is present in the dataset.
- The dataset will be stored on a password-protected Google Drive account accessible only to the researcher.
- The DHS Program data use agreement strictly prohibits redistribution of the dataset or any attempt to re-identify individual records. Full compliance with this agreement will be maintained throughout the study.
- The Claude API receives only computed risk scores, SHAP values, and variable-level summary information — no individual-level KDHS records are transmitted to the Anthropic API.

3.8.3 Research Integrity

- All model results, performance metrics, and analytical outputs will be reported accurately and transparently, including cases where model performance falls below expectations.
- The composite risk variable engineering methodology will be fully documented and disclosed to enable replication.
- All code will be version-controlled on a GitHub repository, which will be made publicly available upon project completion to support open science.
- Authorship and data attribution will comply with DHS Program citation requirements and USIU-Africa academic integrity policies.

3.8.4 Potential Harms and Mitigations

The primary ethical risk of this study is the potential for model misclassification — specifically, the assignment of Low Risk to a genuinely High Risk woman, leading to a missed referral. This risk is mitigated by: (1) prioritising Recall for the High Risk class in model selection; (2) communicating Bayesian uncertainty intervals to CHWs so that borderline classifications are flagged as requiring additional assessment; and (3) clearly positioning the application as a clinical decision support tool that augments rather than replaces CHW clinical judgement. The application interface will include a disclaimer stating that all predictions should be interpreted alongside clinical assessment and the CHW's professional judgement.

3.9 Chapter Summary

This chapter has presented the comprehensive methodological framework for the proposed study. The study adopts a quantitative, non-experimental, secondary data analysis design, drawing on the KDHS 2022 Individual Recode dataset — a nationally representative survey of 17,414 Kenyan

women. The experimental pipeline progresses through five structured phases: data preprocessing, target variable engineering, comparative model development, Bayesian inference integration, and Streamlit application deployment with Claude AI integration.

The model development strategy is distinguished from prior studies by the inclusion of novel deep learning architectures for tabular data — specifically TabNet and TabTransformer — alongside classical ensemble methods, with Optuna-based Bayesian hyperparameter optimisation and a stacking meta-ensemble as the final model architecture. Model evaluation prioritises healthcare-appropriate metrics, with High Risk class Recall and macro AUC-ROC as primary performance indicators. Bayesian inference via PyMC provides calibrated probability estimates with credible intervals, transforming the model's output from a binary classification to a probabilistically grounded, uncertainty-aware risk score.

The methodology is ethically sound, operating exclusively on anonymised secondary data under NACOSTI and IRB oversight and full compliance with the DHS Program data use agreement. The following chapter will present the results of the model training, evaluation, and application deployment phases.

References

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APPENDICES

Appendix A — Introduction / Data Request Letter

> *Template: Present this letter to your supervisor for signature before NACOSTI/IRB submission.*

To Whom It May Concern,

I am Selmah Tzindori Mise, a Senior student in the Bachelor of Science in Data Science and Analytics programme at the United States International University Africa (USIU-Africa), Nairobi, Kenya (ID: 669602). I am writing to request access to the Kenya Demographic and Health Survey 2022 (KDHS 2022) — specifically the Individual Recode (IR) dataset — for the purposes of my final year research project titled 'Predicting Maternal Health Risk Among Pregnant Women in Kenya Using Machine Learning and Bayesian Inference on the KDHS 2022.'

The data will be used exclusively for academic research purposes, stored securely on a password-protected device, and not redistributed. All analysis will comply fully with the DHS Program data use agreement. No attempt will be made to re-identify individual records. I kindly request your approval to proceed with data access under these conditions.

Yours sincerely,

Selmah Tzindori Mise | ID: 669602 | USIU-Africa | [Date]

Supervisor: _____ | Signature: _____

Appendix B — NACOSTI & IRB Ethical Approval Form (Debrief)

This study uses the Kenya Demographic and Health Survey 2022 — a fully anonymised, publicly available secondary dataset. No primary data collection is conducted. No human participants are recruited, interviewed, observed, or contacted. All data are de-identified prior to release by KNBS and ICF International. The research poses no physical, psychological, social, or legal risk to any individual. Ethical approval is sought to formally document compliance with USIU-Africa's research ethics requirements. Data will be analysed solely for academic purposes and destroyed upon project completion.

Appendix C — Work Plan / Timeline

Week	Phase	Key Task	Deliverable
1–2	**Proposal**	Topic + problem statement + literature	Week 2 report
3–4	**Proposal**	Dataset access + methodology draft	Week 4 report
5–6	**Proposal**	Expected results + scope + Gantt chart	Week 6 document
7–8	**Proposal**	Concept note preparation + presentation	Concept note slides

Week	Phase	Key Task	Deliverable
9–10	**Proposal**	Full proposal document + NACOSTI/IRB submission	Proposal document
11–12	**Proposal**	Proposal presentation preparation	Proposal defense
Sem 2 W1–4	**Implementation**	Data preprocessing + EDA + feature engineering	Clean dataset
Sem 2 W4–6	**Implementation**	Baseline model training + evaluation	Model comparison table
Sem 2 W6–9	**Implementation**	TabNet + TabTransformer + stacking ensemble	Best model selected
Sem 2 W9–10	**Implementation**	Bayesian inference + SHAP integration	Probability outputs
Sem 2 W10–12	**Implementation**	Streamlit app + Claude AI integration	Working prototype
Sem 2 W12–14	**Implementation**	Testing + report writing + final defense	Final report

Appendix D — Budget

Item	Purpose	Estimated Cost	Currency
Anthropic Claude API credits	AI explanation layer	$5.00 (one-off top-up)	USD
Google Colab (free tier)	Model training environment	$0.00 (free)	—
Streamlit Community Cloud	App deployment	$0.00 (free)	—
GitHub (free tier)	Code version control	$0.00 (free)	—
DHS Program data access	KDHS 2022 dataset	$0.00 (academic free)	—
Printing & binding (final report)	Physical submission copy	KES 1,500	KES
Internet data (research + uploads)	Literature search + Colab access	KES 2,000	KES
TOTAL		**~KES 3,500 + $5.00**	—

Appendix E — Data Collection Sheet / Variable Codebook

The following variables are extracted from the KDHS 2022 Individual Recode (IR) file using the Python code below. This constitutes the data collection instrument for this study:

```
columns_needed =
['v012','v201','v106','v190','v025','v501','v212','v024','m14_1','m15_1','m17_1','m
3a_1','b11_01']

maternal_df = df[columns_needed].copy()
```

Appendix F — Model Architecture Specifications

The following table summarises the architecture specifications of the novel models proposed in this study:

Parameter	TabNet	TabTransformer
Architecture type	Sequential attention with sparse feature selection	Transformer encoder with contextual categorical embeddings
Input handling	Continuous + categorical (unified)	Categorical features → Transformer; continuous → MLP
Interpretability	Built-in attention masks (step-wise feature importance)	SHAP + column-level attention weights
Hyperparameters (key)	N_a (attention width), N_steps, gamma, lambda_sparse	n_layers, n_heads, dim_head, mlp_dim
Optimisation	Optuna (100 trials, 5-fold CV)	Optuna (100 trials, 5-fold CV)
Python library	pytorch-tabnet	tab_transformer_pytorch
Citation	Arik & Pfister, 2021 (AAAI)	Huang et al., 2020 (arXiv:2012.06678)

Appendix G — Consent and Data Use Agreement

This study operates under the DHS Program Data Use Agreement, accepted by the researcher upon registration at dhsprogram.com. Key terms include: the dataset is for use in the researcher's registered project only; no redistribution of the data to third parties is permitted; findings may be published provided the DHS Program is cited as the data source; and no attempt to re-identify individual respondents will be made. A copy of the signed data use agreement is retained by the researcher and is available for review by the NACOSTI/IRB committee upon request.

Appendix H — Supplementary Analysis Plan

Upon completion of the primary model evaluation, the following supplementary analyses will be conducted and reported: (1) county-level disaggregation of predicted risk scores to generate the 47-county risk dashboard; (2) subgroup analysis by urban/rural residence and wealth quintile to assess model equity across demographic segments; (3) sensitivity analysis varying the risk flag threshold (2 flags vs. 3 flags for High Risk) to assess the robustness of the classification boundary; and (4)

calibration curve analysis comparing the Bayesian probability outputs against empirical event rates in the test set.